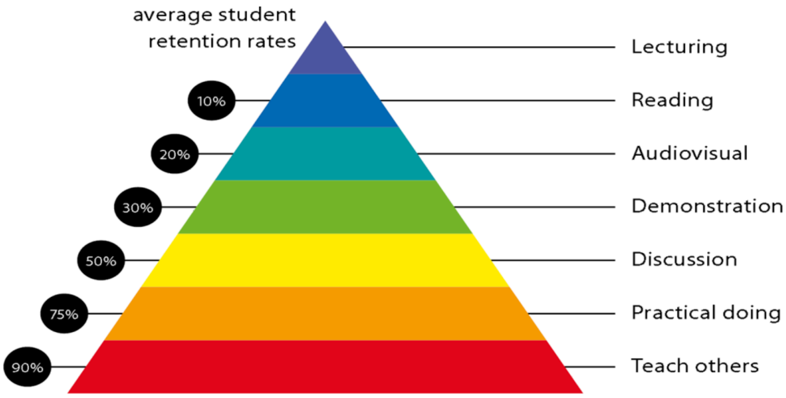

Figure 2: Knowledge retention

Practical work:

Wk No.	Class Activity	List of Practical
01	Lab 1	Write a program to implement following searching algorithms. Linear (2) Binary.
02	Lab 2	Write a program to implement following sorting algorithms. Bubble (2) Selection (3) Quick (4) Merge
03	Lab 3	Write a program to implement following STACK operations. PUSH (2) POP (3) PEEP (4) CHANGE (5) DISPLAY
04	Lab 4	Write a program to convert infix expression to postfix expression.
05	Lab 5	Write a program to implement following QUEUE operations. INSERT (2) DELETE (3) DISPLAY
06	Lab 6	Write a program to implement following CQUEUE operations. INSERT (2) DELETE (3) DISPLAY
07	Lab 7	Write a program to implement following operations of the singly linked list (SLL). (1) Insert a node at the front of the linked list. (2) Insert a node at the end of the linked list. (3) Insert a node such that linked list is in ascending order. (According to info. Field)
08	Lab 8	Write a program to implement following operations of the singly linked list (SLL). (1) Delete a first node of the linked list. (1) Delete a node before specified position. Delete a node after specified position.
09	Lab 9	Write a program to implement following operations of the doubly linked list (DLL). (1) Insert a node at the front of the linked list. Insert a node at the end of the linked list.
10	Lab 10	Write a program to implement following operations of the doubly linked list (DLL). (1) Delete a last node of the linked list. Delete a node before specified position.
11	Lab 11	Write a program to implement stack using linked list.
12	Lab 12	Write a program to implement queue using linked list.

13	Lab 13	Write a program to implement binary tree traversals.
		Practical Beyond syllabus
14	Lab 14	Write a Program to implement heap sort.
15	Lab 15	Write a Program to implement priority queue.

Lecture/tutorial times

(Put your slots as per your time table)

Attendance Requirements

The University norms states that it is the responsibility of students to attend all lectures, tutorials, seminars and practical work as stipulated in the Course outline. Minimum attendance requirement as per university norms is compulsory for being eligible for mid and end semester examinations.

Details of referencing system to be used in written work

Text books:

Text Books

1. An Introduction to data structures with applications. By jean paul tremblay & paul G Sorenson publisher TMH.

Reference Books

1. Data Structures using C & C++ -By Ten Baum Publisher – Prenctice-Hall International.
2. Fundamentals of Data Structures in C++-By Sartaj Sahani.
3. Classical Data Structure by D. samantha. Pearson publication.

Additional Materials

https://www.studytonight.com/data-structures/introduction-to-data-structures
https://www.javatpoint.com/data-structure-tutorial
https://www.tutorialspoint.com/data_structures_algorithms/data_structures_basics.htm

Syllabus

UNIT-I	[12 Hours]
INTRODUCTION TO DATA STRUCTURE: Definition, classification of data structure, Examples of data structure. Searching and Sorting: Various sorting techniques: Selection sort -bubble sort -Quick sort, Merge sorting. Sequential searching, Binary searching.	
UNIT-II	[12 Hours]
LINEAR DATA STRUCTURE:Representation of arrays, Applications of arrays, Stack: Stack-Definitions & Concepts, Operations On Stacks, Applications of Stacks, Polish Expression, Reverse Polish Expression And Their Compilation, Recursion, Tower of	